Jianan Jiang

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Education

- **Hunan University** (GPA: 3.51/4.0), Under Prof. Di Wu *Sep. 2022 - Jun. 2025*
  *M.E. in Design Science (Cross-disciplinary), **Direct admission*** Changsha, China
- **Donghua University** (GPA: 3.62/5.0), Under Prof. Jie Zhang & Dr. Junliang Wang *Sep. 2018 - Jun. 2022*
  *B.E. in Mechanical Engineering (Mechatronics Engineering), **Shanghai Outstanding Graduates** (5%)* Shanghai, China
  - Mechanical Engineering Excellence Class *Sep. 2018 - Jun. 2022*
  - Intelligent Manufacturing and Robotics Experimental Class *Dec. 2019 - Jun. 2022*
  - Artificial Intelligence Experimental Class *Dec. 2020 - Jun. 2022*

Publications

** Equal Contribution*

- *Y. Lu*, X. Shi*, **J. Jiang**, H. Deng, Y. Wang, J. Lu, D. Wu.* "FPGA Adaptive Neural Network Quantization for Adversarial Image Attack Defense." IEEE Transactions on Industrial Informatics (**TII**), 2024.
- ***J. Jiang**, D. Wu, H. Deng, Y. Long, W. Tang, X. Li, C. Liu, Z. Jin, W. Zhang, T. Qi.* "HAIGEN: Towards Human-AI Collaboration for Facilitating Creativity and Style Generation in Fashion Design." The ACM International Joint Conference on Pervasive and Ubiquitous Computing (**Ubicomp**), 2024.
- *D. Wu*, M. Wu*, Y. Li*, **J. Jiang**, X. Li, H. Deng, C. Liu, Y. Li.* "StyleWe: Towards Style Fusion in Generative Fashion Design with Efficient Federated AI." The 27th ACM Conference on Computer-Supported Cooperative Work and Social Computing (**CSCW**), 2024.
- *H. Deng*, **J. Jiang***, Z. Yu, J. Ouyang, D. Wu.* "CrossGAI: A Cross-Device Generative AI Framework for Collaborative Fashion Design." The ACM International Joint Conference on Pervasive and Ubiquitous Computing (**Ubicomp**), 2024.
- *D. Wu*, Z. Yu*, N. Ma*, **J. Jiang**, Y. Wang, G. Zhou, H. Deng, Y. Li.* "StyleMe: Towards Intelligent Fashion Generation with Designer Style." Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (**CHI**), 2023.

- D. Wu, M. Wu, **J. Jiang**, X. Li, H, Deng. "An AIGC federated learning approach for designer style fusion and privacy protection." CN202410389515[P]. 2024-05-30. (CN Patent)
- D. Wu, Z. Yu, **J. Jiang**, X. Li. "An Intelligent Clothing Image Generation System with Designer Style." CN202211571187[P]. 2023-01-17. (CN Patent)
- J. Zhang, J. Wang, **J. Jiang**, J. Guo, D. Li, H. Shen, J. Li, W. Wang, X. Xiao, F. Xu. "A spare yarn winding device for automatic joint of ring spinning yarn." CN202110398137[P]. 2021-11-05. (CN Patent)
- J. Zhang, J. Wang, **J. Jiang**, J. Guo, D. Li, H. Shen, J. Li, W. Wang, X. Xiao, F. Xu. "A Robot End Actuator for Automatic Joint of Ring Spinning Yarn." CN202110398252[P]. 2021-07-27. (CN Patent)
- J. Zhang, J. Wang, D. Li, J. Guo, **J. Jiang**, J. Li, W. Wang, X. Xiao, F. Xu, H. Shen. "An automatic joint device and method of ring spinning spinning yarn machine." CN202110398139[P]. 2021-07-27. (CN Patent)
- 2021SR0953602. "A Robot automated production line big data application platform for the textile production process V1.0." 2021-05-15. (Computer Software Copyright)
- 2021SR1280046. "A Cotton fiber flexible work robot scheduling cloud system V1.0.0." 2021-05-01. (Computer Software Copyright)

Research Projects

- **Intelligent Design and Supply Chain Management for the Customized Chinese Style Clothing** *Main Contribution*
  Key R&D Program of Hunan Province, Under Prof. Di Wu Jan. 2023 - Dec. 2023
  - **Introduction**: Data modeling and style transfer of Chinese style clothing.
  - **Responsibility**: Text-to-image model fine-tuning (Stable Diffusion), personalized sketch generation (Lightweight GAN), sketch-to-image style transfer (DDIM).
- **5G and Edge Computing Driven Human-Computer Interaction for Product Design** *Main Contribution*
  National Key R&D Program of China, Under Prof. Di Wu Aug. 2022 - Dec. 2022
  - **Introduction**: Design and application of multi-terminal human-computer interaction system for clothing products based on 5G and edge computing.
  - **Responsibility**: Image Style Transfer Based on Sketch (AE & GAN), Image Retrieval Based on Fine-grained Sketch (ViT & Contrast Learning).
- **Broken Yarn Joint Robot** *Team Lead*
  National Key R&D Program of China, Under Prof. Jie Zhang & Dr. Junliang Wang Nov. 2020 - May. 2022
  - **Introduction**: China is the largest textile country, and the broken yarn automatic joint technology is the most important part of the intelligent transformation of textile factories. This project designs a system of collaborative operation between inspection robots and joint robots to realize automatic broken yarn detection and joint.
  - **Responsibility**: (i) In terms of hardware, Solidworks and 3D printing technology are combined to design and build inspection and joint robot entities. (ii) In terms of software, a "cloud-edge" architecture based on Socket communication is designed, a cloud monitoring system based on the Vue framework is built, and realized: edge yarn break detection based on Jetson Nano and YOLO-V5, edge joint robot precision positioning based on Hikvision industrial camera and MVS application, edge joint robot data processing based on Raspberry Pi, UR six-axis robot and other execution devices controlling, etc.

- **An Intelligent Control System for College Student Vehicle Competition** *Project Lead*

National Innovation and Entrepreneurship Training Programme for University Students Sep. 2020 - Aug. 2021

  - **Introduction**: An intelligent control system is designed for the problems of uncentralized and unshared driving data in college student vehicle competitions.
  - **Responsibility**: Multi-sensor signal acquisition and processing based on Stm32, using GPS module and other sensors to communicate with off-site personnel in real-time and share pictures.

- **A Clothes Drying System Based on Weather Detection** *Project Lead*

Shanghai Innovation and Entrepreneurship Training Programme for University Students Sep. 2020 - Aug. 2021

  - **Introduction**: Aiming at the sudden bad weather and other problems encountered in the process of drying clothes, a clothes drying and drying system for indoor balconies is designed.
  - **Responsibility**: Based on the collaborative operation of multiple sensors and actuators based on Stm32, a multi-application real-time monitoring system based on the TLINK Internet of Things platform is built.

- **Buyue Energy-saving Vehicle Team of Donghua University** *Project Lead*

University Student Association, Team Leader Nov. 2019 - Apr. 2021

  - **Introduction**: Participate in the Honda China Energy Conservation Competition to achieve longer vehicle driving distance with minimal energy (oil or electricity) consumption.
  - **Responsibility**: The overall design of the energy-saving vehicle scheme and the production and debugging of the whole vehicle, including the design of the car body structure and car body based on SolidWorks, the analysis and optimization of the car body drag coefficient based on Ansys, the modification and weight reduction of the horizontal bar engine, and the handmade carbon fiber car body, etc.

- **Yarn Quality Inspection** *Participate*

Under Prof. Jie Zhang & Dr. Junliang Wang Nov. 2019 - Feb. 2020

  - **Introduction**: The online inspection of yarn quality in the textile field has important engineering value for leveling the whole dry, optimizing hairiness, and reducing breakage.
  - **Responsibility**: The design and construction of experimental equipment, learning the object detection algorithm based on Faster R-CNN.

- **Fabric Defect Detection** *Participate*

Under Prof. Jie Zhang & Dr. Junliang Wang Sep. 2019 - Nov. 2019

  - **Introduction**: In the textile field, human eye inspection of the quality of grey fabrics will seriously affect human health. This project has designed a multi-category grey fabric defect detection equipment.
  - **Responsibility**: The design and construction of the experimental equipment, using the collaborative work of multiple motors in the PLC control equipment, learning the object detection algorithm based on YOLO-V3.

Technical Skills

- **Programming Languages**: Python/Pytorch/Tensorflow/Keras, C/C++, HTML/CSS, etc.
- **Tools and Frameworks**: Windows/Ubuntu, Jetson Nano, Raspberry Pi, Stm32, PLC, Solidworks, Altium Designer, Ansys, etc.

Selected Awards

- **National Second Prize**, The 17th "Challenge Cup" National Undergraduate Extracurricular Academic Science and Technology Works Competition (Top1 Competition in China) *Mar. 2022*
- **Shanghai First Prize**, The 17th "Challenge Cup" Shanghai Undergraduate Extracurricular Academic Science and Technology Works Competition *May. 2021*
- **Shanghai Third Prize**, The "Higher Education Cup" National College Students Mathematical Contest in Modeling *Dec. 2020*
- **Shanghai Second Prize**, The 10th Advanced Mapping Technology and Innovative Design Competition *Dec. 2020*
- **National Third Prize**, The China Undergraduate Mechanical Engineering Innovation and Creativity Competition — The 4th China Intelligent Manufacturing Competition *Oct. 2020*
- **Nationwide 10th**, The 13th Honda China Energy-saving Competition (Fuel Group) *Nov. 2019*
- **Nationwide 9th**, The 13th Honda China Energy-saving Competition (EV Group) *Nov. 2019*
- **Shanghai First Prize**, The "Hirschvogel Cup" Shanghai Undergraduate Creative Robot Challenge Competition *Oct. 2019*
- **National Third Prize**, The 2019 "Robocup" China Robot Competition *Aug. 2019*
- **National First Prize**, The 6th Educational Robot Competition in China *Nov. 2016*